

Agent under study

Evaluation harness

Inputs

TARGET FUNCTION
or clone under analysis

VULNERABILITY
KNOWLEDGE BASE
CVE records, known fixes

*pre/post-cutoff CVEs,
certified variants*

LLM DETECTOR
vulnerable or safe?

PROGRAM SLICER
vulnerability slice (srcSlice)

CVE TEXT
description, CWE

KNOWN-FIX RETRIEVAL
srcVul slicing vectors

grounding sources

LLM REPAIR AGENT
invents a fix, or
ports a known fix
returns the complete
patched function

PATCHED FUNCTION

SELF-CHECKS
static checks; slice severance

PAIR-CORRECTNESS
verdicts on vulnerable
and fixed versions

INDEPENDENT ORACLE
agrees with the
official fix? → VPR

OFFICIAL FIXES
and labels

only place the ground truth enters

verdict

candidate

feedback

Data & inputs
Slicing / static analysis
LLM component
Evaluation
→ grounding
 - - -> feedback
 ⋯→ ground truth
 RQ probed by research question